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Affective speech modulates a cortico-limbic network in real time

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ABSTRACT

Affect signaling in human communication involves cortico-limbic brain systems for affect information decoding, such as expressed in vocal intonations during affective speech. Both, the affecto-acoustic speech profile of speakers and the cortico-limbic affect recognition network of listeners were previously identified using non-social and non-adaptive research protocols. However, these protocols neglected the inherent socio-dyadic nature of affective communication, thus underestimating the real-time adaptive dynamics of affective speech that maximize listeners' neural effects and affect recognition. To approximate this socio-adaptive and neural context of affective communication, we used an innovative real-time neuroimaging setup that linked speakers' live affective speech production with listeners' limbic brain signals that served as a proxy for affect recognition. We show that affective speech communication is acoustically more distinctive, adaptive, and individualized in a live adaptive setting and more efficiently capitalizes on neural affect decoding mechanisms in limbic and associated networks than non-adaptive affective speech communication. Only live affective speech produced in adaption to listeners' limbic signals was closely linked to their emotion recognition as quantified by speakers' acoustics and listeners' emotional rating correlations. Furthermore, while live and adaptive aggressive speaking directly modulated limbic activity in listeners, joyful speaking modulated limbic activity in connection with the ventral striatum that is, amongst others, involved in the processing of pleasure. Thus, evolved neural mechanisms for affect decoding seem largely optimized for interactive and individually adaptive communicative contexts.

1. Introduction

Previous research described the cortical and subcortical brain systems that recognize affective signals expressed by other individuals (LeDoux, 2012; Pessoa, 2017) with common and specific brain systems for the recognition of auditory emotions expressed in conspecific vocalizations and affective speech (Frühholz et al., 2016b; Frühholz and Staib, 2017; Scott et al., 1997). A central affect recognition system is the limbic system, with a specific role of the amygdala (Fecteau et al., 2007; LeDoux, 2012), which often functions in collaboration with the auditory cortex (AC) (Ethofer et al., 2009; Frühholz et al., 2015a) and other (sub-) cortical brain systems (Frühholz et al., 2016b; Pessoa, 2017). Reports about the involvement of the amygdala in affective speech recognition are however inconsistent (Bestelmeyer et al., 2014; Frühholz et al., 2016b, 2014; Wiethoff et al., 2009). One explanation for this inconsistency might be the use of non-social and non-adaptive research protocols, especially the use of pre-recorded and thus non-adaptive affective speech presented to listeners. Given that affective communication between a speaker and a listener is highly dynamic and adaptive, and given that the amygdala was evolutionarily shaped in dynamic and real-time (social) interactions with the environment (Janak and Tye, 2015; LeDoux, 2012), previous research might not have captured the full and consistent processing capacities of the limbic system for affective speech recognition.

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We here introduced an innovative experimental design for real-time neuroimaging that included both a human speaker and a listener who were connected in a closed-loop setup (Fig. 1). This setup allowed speakers to dynamically modulate their affective speech in response to amygdala signal that was fed back in real time from the listeners' brains. The amygdala activity was presented to the speakers by visual analog feedback in the form of a continuously changing bar graph. The speakers were actors whom we trained to introduce vocal variation by modulating their voices in different intonation features. Speakers could thus optimize their affective speech by adapting various vocal features in order to dynamically maximize neural effects in the limbic system of listeners (which served as an indicator for optimizing listeners' emotional understanding). Using functional magnetic resonance imaging (fMRI), we recorded both whole brain and left amygdala functional brain activity in human listeners ($n = 24$, 14 females, 10 males); The left amygdala was chosen as the target region for quantifying the neurofeedback signal, given its more consistent involvement in affective speech recognition (Frühholz et al., 2015a; Frühholz and Grandjean, 2013). Listeners perceived affective speech by speakers ($n = 8$, 4 females, 4 males) who expressed either aggression or joy, which are among the most important emotions for social interactions (Grandjean et al., 2005; Scott et al., 2014). The listeners' task was to listen passively but attentively to the speaker's vocalizations. Actors produced affective speech in two speaking modes: a pre-recorded condition (which we refer to as the "recorded" condition), as commonly used in many previous studies (Bestelmeyer et al., 2014; Ethofer et al., 2009; Frühholz et al., 2015a; Grandjean et al., 2005), and an innovative real-time condition (which we refer to as the "live" condition). While the live speech condition allowed speakers to continuously adapt their affective speech in response to brain signals from listeners, the recorded condition included no such individual adaptions and is therefore considered a non-adaptive speaking condition. After the experiment, the listeners gave perceptual ratings of each affective intonation. This setup allowed us to quantify communication-related signals on three different levels: on the listeners' side, we quantified the (a) neural (i.e., amygdala, whole brain, and network) and (b) perceptual dynamics (i.e., valence and arousal ratings) of perceiving the speakers' affective speech intonations; and (c) on the speakers' side, we quantified the acoustic profile of the affective speech. Additionally, to examine communication from an interactive expression-perception perspective, we linked the speakers' acoustic speech profiles with the listeners' neuro-perceptual effects. By comparing the different types of signals measured during the live and the recorded condition, we aimed to quantify significant differences between both conditions on the neural, perceptual, and acoustic level as well as in the speaker-listener coupling between those levels during affective speech communication.

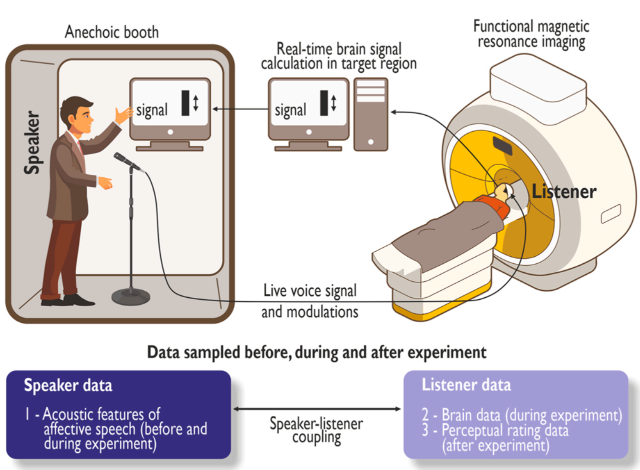

Fig. 1. Closed-loop setup linking a speaker and listener in adaptive affective speech communication. Speakers continuously modulated their affective speech in response to the amygdala signal quantified in listeners' brains while they listened to the speakers' voices. The amygdala signal was quantified in real time and visually displayed to speakers as a continuously updated bar graph representing the percentage signal change at each scanning time-point. The speakers' task was to modulate their affective speech intonations with the aim to maximize the amygdala signal in listeners. Experimental data were sampled on three levels and at different time-points regarding the experiment: (1) speakers' affective speech was recorded in the pre- and live session of the experiment, (2) listeners' whole brain and amygdala signals were quantified during the experiment, and (3) listeners provided post-experimental emotion ratings on the speakers' speech intonations.

2. Results

2.1. Real-time affective speech drives dynamic neural activity and connectivity patterns

To assess the effects of the dynamically expressed affective speech, we quantified the neural effects in listeners elicited by the live affective speech performance of the speakers. We were specifically interested in neural effects in the amygdala and the AC (Frühholz et al., 2015a; Pannese et al., 2016), as well as in associated areas in the striatum and the inferior frontal cortex belonging to the extended voice processing network (Frühholz et al., 2016b) (all functional images were thresholded with voxel-level $p < 0.005$ and cluster extent $k > 46$). Concerning the AC, we first determined the extent of the general voice area (VA) that is sensitive to voice and speech signals by using a standard functional localizer scan (Pernet et al., 2015) (Fig. 2). The VA covered large parts of the bilateral AC, with peak locations in the mid-superior part of the higher-order AC labeled as Te3. This VA definition served to identify if significant clusters of activity in the main affective speech experiment belonged to a voice-sensitive region.

Comparing live versus recorded speech across both emotions (Fig. 3A), we found activity peaks in the bilateral AC (located within the VA on the higher-order superior temporal cortex (STC) and sulcus (STS), and the low-level Heschl's gyrus), right frontal cortex (premotor cortex (PMC), frontal operculum (fOP)) and intraparietal sulcus (IPS), and left striatal areas (dorsal striatum (dStr), ventral striatum (vStr)). Most importantly, we found increased activity in the left amygdala, the target region of the real-time neurofeedback signal displayed to speakers. No regions showed higher activation during recorded compared to live speech. The MNI (Montreal Neurological Institute) coordinates of all significant peak voxels in all contrasts (within and across emotions) are listed in *SI Appendix*, Table S1.

Comparing the live against the recorded condition for each emotion separately (Fig. 3B), we found cortical activity in the right AC and subcortical activity in the bilateral amygdalae for aggressive speech. For joyful speech, the same contrast revealed cortical activity in the bilateral AC, the right PMC, fOP, and IPS, and subcortical activity in the bilateral vStr and dStr. Thus, the activity elicited by live joyful speech was spread over multiple cortical areas in the right hemisphere and also elicited activity in the left AC, which was not the case in aggressive speech. Both live affective conditions induced cortical activity in the right AC but differential activity in subcortical areas. During live aggression, there was limbic activity, particularly in the left amygdala and to a lesser extent also in the right amygdala, but there was no striatal activity. Live joy showed mainly subcortical activity in the bilateral vStr and dStr but not in the amygdala. While, when looking at the beta estimates extracted from the amygdala (Fig. 3C), the amygdala activity seems to be higher also in live joyful voices, this difference was not significant. Therefore, it seemed that live joyful speech did not directly modulate left amygdala activity. However, a functional connectivity analysis showed a significant involvement and modulation of the left amygdala in a broader cortico-striato-limbic network.

To examine the network involved in live dynamic and non-adaptive affective speech processing, we used Granger causality analysis (GCA) to

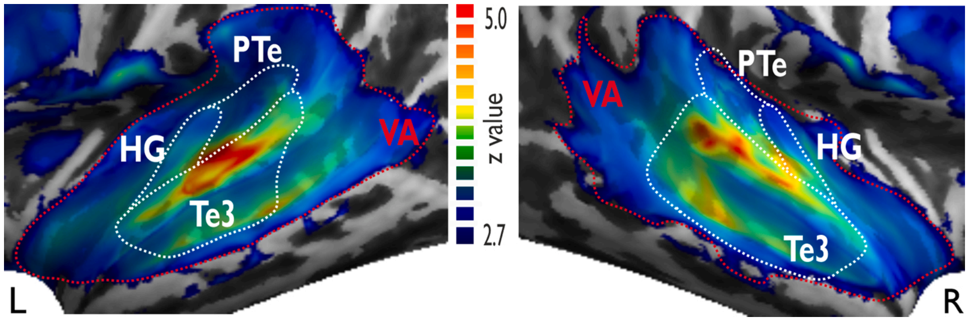

Fig. 2. Neural voice area as defined by the functional voice localizer. Red dashed line delineates boundaries of the VA ($p < 0.005$, cluster size $k > 46$ voxels, corrected on the cluster level) and white dashed line shows localization of anatomical subregions of the AC.

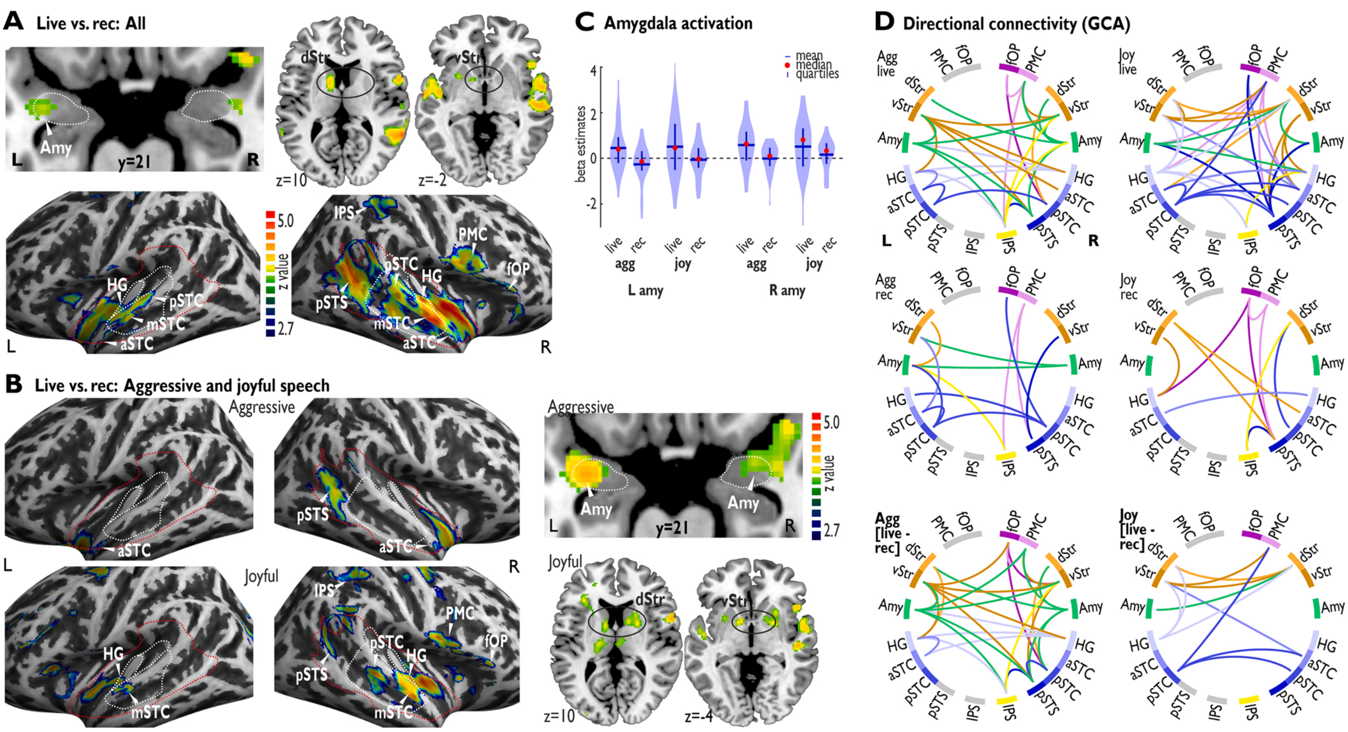

Fig. 3. Real-time affective speech drives neural activity and directed connectivity. (A) Neural activations ($p < 0.005$, $k > 46$) during live versus recorded speech across both emotions and (B) specifically for aggression (upper panel) and joy (lower panel). The left and middle panels show cortical activations, and the right panel shows subcortical activations. Live aggressive speech elicited subcortical activation only in the amygdala and joyful speech only in the striatum. Red dashed line delineates VA and white dashed line anatomical subregions of the AC. (C) Extracted beta values representing the amygdala responses. (D) Functional connectivity analysis (Granger causality analysis, GCA; $p < 0.01$, FDR-corrected) for each of the four conditions (upper and middle panel) and for the difference between live and recorded speech (lower panel); color of connections is defined by the region of origin. Abbreviations: *VA* voice area, *AC* auditory cortex, *FDR* false discovery rate, *rec* recorded, *Agg* aggressive, *L/R* left/right hemisphere, *Amy* amygdala, *dStr/vStr* dorsal/ventral striatum, *PMC* premotor cortex, *fOP* frontal operculum, *HG* Heschl's gyrus, *aSTC/mSTC/pSTC* anterior/mid/posterior superior temporal cortex, *pSTS* posterior superior temporal sulcus, *IPS* intraparietal sulcus.

determine directional neural network connectivity during the perception of live and recorded speech, separately for aggressive and joyful intonations (Fig. 3D). We entered regions originating from the general live against recorded speech contrasts into the GCA (see *SI Appendix*, Table S1 for the MNI coordinates of the peak voxels of each region of interest (ROI)). During both live and recorded speech, we found an extended neural network between our regions of interest for both emotions (F-statistics, $p < 0.01$, false discovery rate (FDR) corrected). Notably, the left amygdala was the origin of many functional connections to other brain areas. This was especially confirmed by significant connectivity difference patterns for live against recorded speech in each emotion. For live minus recorded aggression, the bilateral amygdalae were part of a highly integrated network and had predominately outgoing connections to many regions. The left amygdala showed connections to the right IPS, AC, and frontal regions, as well as to the bilateral vStr, while only receiving input from the right Heschl's gyrus (i.e., primary AC). The right amygdala showed a similar connectivity pattern but connected to the dStr instead of the vStr. While for the left amygdala, all connections but one were to the contralateral hemisphere, for the right amygdala, all connections but one were to the ipsilateral hemisphere. The regions that received input from both amygdalae were further highly interconnected with each other. For live minus recorded joyful speech, we found a less interconnected network but still an integration of the left amygdala in a wider neural circuitry linking it to the right dStr, which was in turn reciprocally connected to the left dStr, which received input from the left Heschl's gyrus. The latter also showed outgoing connections to the right dStr and to the left vStr. Additionally, the left posterior higher-order AC (pSTC) was connected to the right higher-order AC, which was also connected to the left vStr. *SI Appendix*, Table S2 provides a full overview of the connectivity parameters between all ROIs.

2.2. Perception of emotional intensity and quality

Given that live affective speech more strongly engaged a cortico-striato-limbic network, we next assessed the emotional intensity and affective quality of these live compared with recorded affective speech intonations as perceived by listeners. After the neuroimaging experiment, listeners dynamically and continuously rated each affective speech period they had heard during the experiment on an arousal (i.e., how intense is the listener's emotional response to the speech condition? How emotionally engaging is the speech?) and valence dimension (i.e., how much is the perceived speech condition of negative or positive affective quality?) (*SI Appendix*, Fig. S1). Speech periods had a duration of 30 s, and we quantified the mean ratings over the 4–30-s interval (Fig. 4A). We included neutral speech from the speakers' pre-recordings as a reference condition.

Valence ratings differed between speech conditions and male and female speakers (two-way repeated measures analysis of variance (rmANOVA), $\alpha = 0.05$, $n = 24$, Greenhouse-Geisser (GG) adjusted p-values, main effect condition: $F_{4,92} = 146.28$, $p = 1.4 \times 10^{-20}$, $\eta^2 = 0.802$), with aggressive being more negative than neutral speech (post hoc test, $\alpha = 0.05$, Bonferroni-corrected, all $ps < 2.2 \times 10^{-8}$) and joyful speech being more positive (all $ps < 9.5 \times 10^{-4}$). Live and recorded affective speech was of comparable emotional quality (all $ps > 0.998$). Within conditions, male and female speech was rated slightly different (interaction effect condition*sex: $F_{4,92} = 4.61$, $p = 0.008$, $\eta^2 = 0.007$), such that female live aggressive speech was rated more negatively than its recorded version ($p = 0.021$). Additionally, listeners perceived male recorded aggression as more negative than that of females ($p = 0.005$).

Arousal ratings quantified the intensity of emotional responding in listeners. Arousal ratings were overall different (rmANOVA, $n = 24$, GG adjusted p-values, main effect condition: $F_{4,92} = 111.11$, $p = 6.7 \times 10^{-16}$, $\eta^2 = 0.718$), with affective speech being more arousing than neutral speech (all $ps < 2.2 \times 10^{-8}$). Within affective speech, listeners perceived live joy as more arousing than was recorded joy

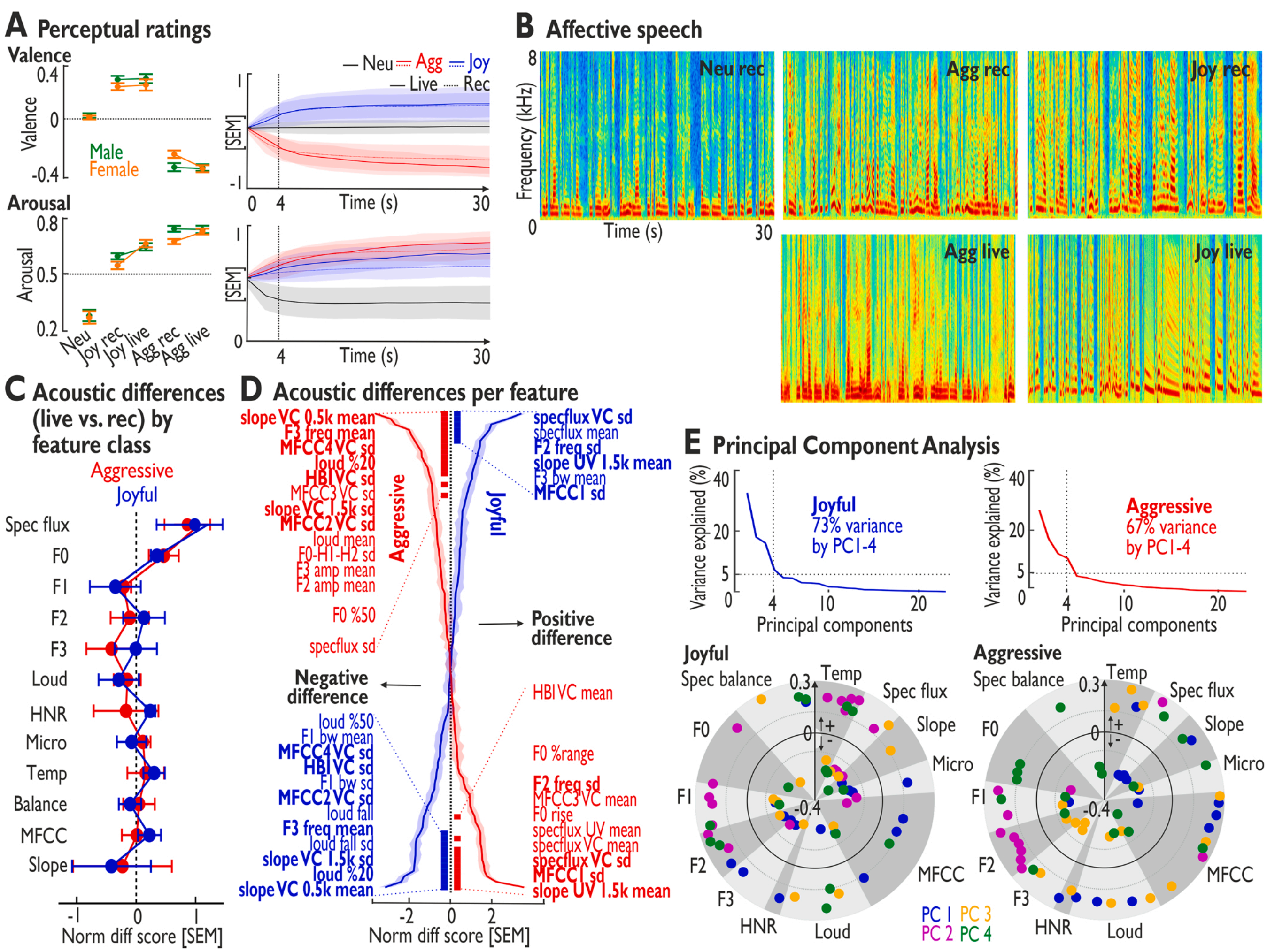

Fig. 4. Analyses of listeners' perceptual ratings and speakers' acoustic data. (A) Mean (+/− 1 SEM) perceptual arousal and valence ratings averaged over the 4- to 30-s speech periods (left panel; male and female voices separated) from continuous ratings (right panels). See also *SI Appendix*, Fig. S1. (B) Spectrograms of example vocalizations for aggressive/agg, joyful/joy, or neutral/neu speech (live or recorded/rec). (C) Normalized difference scores for 88 acoustic vocal features (live minus recorded) summarized by acoustic feature class. (D) Normalized difference scores for each of the 88 vocal features. Features are sorted from maximum positive to maximum negative differences (joy) or vice versa (aggression); thick bars indicate significant differences (two-sided one-sample t-tests, $p < 0.0001$, FDR-corrected) for the respective features, and a bold print of features indicates a correspondence between both emotions. (E) PCA across acoustic difference scores for aggressive and joyful speech. Profile plots of factor loadings for PCs 1–4 for joy and aggression sorted by major acoustic feature classes (lower panel); only loadings beyond +/− 1.5 SD are shown. See also *SI Appendix*, Fig. S2 for more detailed profile plots. Abbreviations: *VC/UV* voiced/unvoiced segments, *F0* fundamental frequency, *F1–F3* formants 1–3, *F0-H1-H2* harmonic difference, *MFCC* Mel-frequency cepstral coefficients, *HBI* Hammarberg Index, *loud* loudness, *specflux* spectral flux, *amp* amplitude, *bw* bandwidth, *freq* frequency, *HNR* harmonics-to-noise ratio, *PC* principal component.

($p = 1.9 \times 10^{-6}$), which was not the case for aggressive speech ($p = 0.450$). Moreover, live and recorded aggressive speech was more arousing than live ($p = 0.002$) and recorded joyful speech ($p = 1.7 \times 10^{-8}$), respectively. Regarding sex differences, affective speech expressed by male speakers was rated higher in arousal than was that expressed by female speakers (main effect sex: $F_{1,23} = 5.711$, $p = 0.026$, $\eta^2 = 0.003$).

2.3. Acoustic nature of real-time and adaptive affective speech

The arousal and valence ratings concerned the perceptual effects in listeners in response to the speakers' affective speech intonations (Fig. 4B). For the latter, we next determined the acoustic profile of live compared with recorded speech. Therefore, we quantified 88 central vocal features commonly found in affective speech communication (Eyben et al., 2016) in terms of vocal pitch, intensity or loudness, and other spectral and temporal features (see *SI Appendix*, Fig. S2 or Table S3 for a full list of all features). To compare live against recorded speech properties, we calculated difference scores (live minus recorded) on which we based all following analyses. Over the broad acoustic feature classes, spectral flux parameters, which are related to the variation of the voice spectral profile, were most influential during live speech, followed by features related to vocal pitch, representing an important cue to the affective meaning expressed in speech intonations (Fig. 4C).

Specifically, we found that both live aggression and live joy were significantly higher on specific vocal features than the respective recorded speech intonations (two-sided one-sample t-tests, difference score between live and recorded against zero, n = 24, $p < 0.0001$, FDR-corrected), especially features related to vocal timbre (low-order Mel-frequency cepstral coefficients (MFCCs)) and spectral voice variations (spectral flux, spectral slope) (Fig. 4D). Live aggressive speech additionally had a higher vocal pitch range and pitch rise and more vocal power in high frequencies (Hammarberg Index of voiced segments). Live joyful speech specifically had increased bandwidth of the third voice formant (F3) (Fig. 4D). In addition to these increased features, live affective speech also showed a significant decrease in certain vocal features, such as mean loudness parameters and certain higher-order MFCCs. Live aggressive speech showed an additional decrease in spectral flux variability and formant-related parameters, while live joyful speech showed specific decreases for first voice formant (F1) parameters and smoother vocal loudness falls (Fig. 4D).

Besides this overall acoustic difference of live versus recorded affective speech, there might also be specific and differential voice modulations introduced by speakers to increase amygdala activity in each listener individually. Some specific voice modulations might elicit a strong amygdala response in one listener, while another set of voice modulations might be successful in another listener. Therefore, we established differential acoustic voice profiles by using a principal component analysis based on the acoustic difference profiles for the 88 features of speakers' voice profiles that were used for each listener individually (PCA; n = 24; Fig. 4E; *SI Appendix*, Fig. S2). This analysis identified four principal vocal profiles separately for aggressive and joyful speech, and each acoustic profile explained > 5% of the overall variance. For example, the first principal acoustic profile for joyful speech showed differential loadings on F2–F3, micro-irregularity, and MFCC parameters, while the second principal acoustic profile showed differential loadings on F0–F2, temporal change, and spectral flux parameters. Thus, we termed the first component a "jittering timbre" vocal profile and the second component a "pitch-and-change" profile. Components 3 and 4 were termed the "loudness" and "articulatory" vocal profiles, respectively, given their major feature loadings. For aggressive speech, we found different principal acoustic profiles. The first principal acoustic profile, for example, showed high loadings on harmonics-to-noise ratio, loudness, MFCC parameters, and some temporal and slope parameters, and largely negative loadings on spectral flux. In contrast, the second principal acoustic profile showed high loadings on F1–F2 and some MFCC and spectral flux parameters. We thus termed the four profiles "loud-and-rough", "distinctive tone", "slow-and-fast change", and "shimmering pitch" vocal profiles for aggressive speech.

2.4. Communicative coupling of speakers' voice features with neural and perceptual listener dynamics

Vocal communication is inherently dyadic and should thus show a coupling of speaker and listener parameters. Therefore, we performed three separate correlation analyses between the speaker and listener data (empirical correlation coefficient distributions compared with null distribution (determined by permutation sampling) using two-sided Kolmogorov-Smirnov tests, $p < 10^{-11}$, FDR-corrected) (Fig. 5; *SI Appendix*, Fig. S3). First, we correlated speakers' acoustic vocal features with listeners' brain activity in selected regions (Fig. 5B). Only for the left posterior STC (pSTC) activity in listeners did we find correlations with acoustic features of the speakers' affective intonations. We found positive correlations with the duration of unvoiced segments (UV mean) for recorded speech and negative correlations with the durational variability of these unvoiced segments (UV variation) for live speech. In addition, we found specific positive correlations with the left pSTC for F2/F3 amplitude and vocal loudness for live aggressive speech.

Second and third, we correlated the acoustic features of the speakers' speech intonations with the listeners' arousal and valence ratings (Fig. 5C). Certain live vocal features were negatively (F1–F3 amplitude variation, UV mean and variation, MFCCs) and positively (F1–F3 amplitude mean) correlated with arousal ratings, both for joyful and aggressive speech, with some specific negative correlations for live joyful (loudness change) and live aggressive speech (spectral flux change). Similar and other features correlated negatively with negative valence ratings for live aggression and positively with positive valence ratings for live joy (F1–F3 amplitude mean, loudness parameters, spectral flux). The valence of live aggression was also negatively associated with voiced segment mean and variation, additional loudness parameters, and spectral flux mean of voiced and UV segments. Live joy additionally correlated with the frequency variation of the second voice formant (F2). Because on the valence dimension a value is more significant the further away from zero it is in both the positive and negative direction, correlations for the negative valence ratings must be interpreted inversely. This means a negative correlation for negative valence ratings equals a positive correlation for positive valence ratings regarding interpretation (i.e., the higher the magnitude of the valence rating in either direction, the higher the brain signal). Reverse correlation patterns of acoustic features with valence ratings (i.e., the higher the magnitude of the valence rating in either direction, the lower the brain signal) were found for some features in both live affective conditions (F1–F3 amplitude variation, loudness variation, UV mean and variation, MFCCs, spectral flux variation). Additionally, in the live aggressive condition, correlations with the alpha ratio variation of voiced segments were found. For recorded speech, we found only some of the correlations found for live speech.

3. Discussion

We used an innovative experimental design for real-time neuroimaging in a closed-loop setup, including speakers who dynamically modulated their affective speech intonations in response to amygdala signal quantified from the listeners' brains. Speakers were asked to optimize their affective speech with the aim to maximize neural effects in the limbic system of listeners and thus to dynamically increase neural processes related to emotion decoding from speakers' speech signals.

3.1. Real-time affective speech drives dynamic neural activity and connectivity patterns

Using this innovative real-time imaging paradigm, speakers

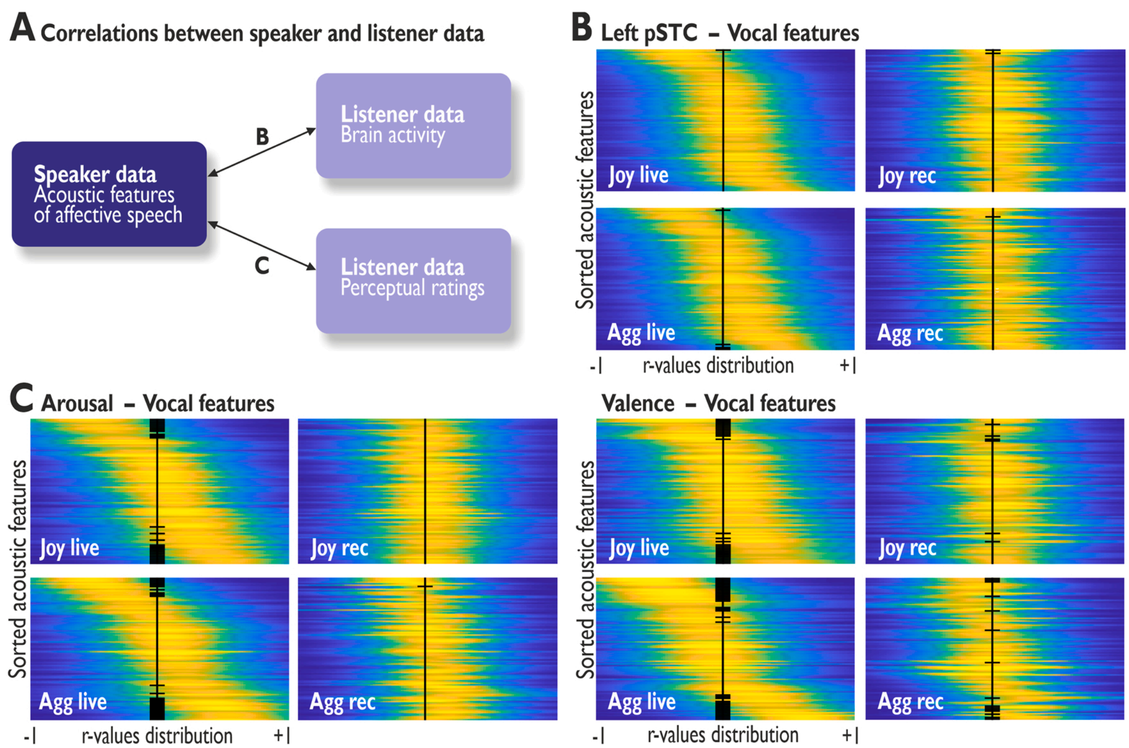

Fig. 5. Correlation analyses between speaker and listener data. (A) Speaker and listener data were correlated in several analyses; letters b and c denote the coupling of specific data represented in plots B and C. (B–C) Density distributions (x-axis: yellow/high density, blue/low density) of Pearson correlation coefficients (each line/ distribution n = 288) between the 88 acoustic features (y-axis) and neural activity in the left posterior superior temporal cortex (pSTC, shown in B) or perceptual ratings (shown in C). Features are sorted according to the mean of the distribution during the corresponding live condition from negative to positive values. Black thick bars in the middle of the plot show a significant difference of the distribution from the null distribution (two-sided Kolmogorov-Smirnov test, $p < 10^{-11}$, FDR-corrected). See also *SI Appendix*, Fig. S3 for more detailed plots.

successfully induced a higher amygdala signal during the live compared to the recorded condition in listeners, which points to the notion that the amygdala seems especially shaped for decoding dynamic and adaptive voice information originating in interactive settings. As a central part of the limbic system, the amygdala's involvement in social and affective processing is widely documented. However, previous reports about its role in affective recognition from sound signals, especially from affective speech, show an inconsistent picture (Frühholz et al., 2016b, 2014). These previous inconsistent findings might be explained by the non-social and non-adaptive research protocols used so far in affective speech stimulation studies. In a real-life context, communication in general and especially affective communication is a highly dynamic and adaptive process between a speaker and a listener. Such real-time social and dynamic interactions evolutionarily shaped the amygdala to be able to react dynamically and adaptively (Janak and Tye, 2015; LeDoux, 2012). Thus, only individually adaptive affective speech might trigger the full and consistent processing capacities of the limbic system involved in real-life affective communication and emotional responding. Therefore, using adaptive speech in our study might have facilitated a more consistent, and thus more robust, amygdala activity in listeners. So, our data demonstrate that, first, speakers adaptively modulate their affective speech in live settings and, second, maximizing amygdala activity in listeners might be a proxy to maximizing their emotional understanding, which is often an important communicative target in daily communication (e.g., increasing aggressive vocal intonations to maximize the intimidating effect in listeners).

Regarding cortical activity, both affective tones elicited similar activations in the right AC, where cortical regions commonly involved in affect decoding from sound are located (Frühholz et al., 2016b). These higher-order areas on the STC and in the STS are generally acknowledged as having a crucial role in affective speech processing (Ethofer et al., 2009; Frühholz et al., 2016b). Especially the right AC seems to be specialized to recognize spectral vocal features (Kotz and Schwartze, 2010), which were among the features with the most pronounced differences between live and recorded speech in our study. Although the right AC thus seems sensitive to emotional voice features, we did not find direct correlations between live speech features and right AC activity. It might be that the right AC decodes acoustic features on a broader time scale to analyze spectral properties (Flinker et al., 2019), probably in terms of general voice impressions that the correlation analysis cannot directly capture. In addition to the right AC's cortical activity, live joyful speech showed more extended cortical activity in right-hemispheric frontal and parietal parts. While the regions in the intraparietal sulcus are part of an important dorsal auditory processing stream (Rauschecker and Scott, 2009), the frontal areas are presumably involved in response preparation (i.e., PMC) and social categorization of sounds (i.e., fOP) (Frühholz et al., 2016b).

Regarding subcortical regions, we found differential activity patterns within the neural affect processing network (Pessoa, 2017). Live aggressive speech directly stimulated the limbic system, especially the left amygdala. Additionally, the left and right amygdala were highly interconnected and showed multiple outgoing connections to other cortico-striatal areas. Unlike in live aggression, subcortical activity during joyful speech was centered on the dorsal and ventral striatum. The vStr is part of the limbic striatum and presumably decodes the pleasantness of sounds (Calder et al., 2004; Zatorrea and Salimpoor, 2013), while the dStr is involved in temporal sound decoding and tracking, potentially detecting relevant vocal changes during dynamic speech (Kotz and Schwartze, 2010). Thus, we did not find a direct modulation of left amygdala activity for live joyful speech. However, the left amygdala was significantly involved and modulated in a broader functional cortico-striato-limbic network, involving both the vStr and the dStr, which was not the case during recorded speech.

The comparison of the networks during the live against the recorded speech intonations showed some differences between the two speech conditions. First, compared to recorded speech, the listeners' brains seemed much more interconnected during the perception of live speech, given a generally increased number of connections between brain regions, specifically also involving the amygdala. Second, there was not only higher interconnectivity during live speech but especially between contralateral regions. Therefore, live speech appeared to stimulate the (interhemispheric) information exchange much more for both emotions.

3.2. Perception of emotional intensity and quality

To extend the level of information about listeners' responses beyond their neural signals, we also quantified their subjective perceptual ratings to the live and recorded speech intonations post-experimentally. We assessed the emotional intensity (i.e., arousal) and affective quality (i.e., valence) of all speech periods as perceived by the listeners. Confirming a different emotional quality and a higher engagement with affective speech, aggressive and joyful speech were rated higher on the valence and arousal scale compared to neutral speech. Furthermore, listeners perceived live joy as more arousing than recorded joy (which was not the case for aggression) and aggressive speech as more arousing than joyful speech. While we expected that live speech intonations would elicit higher arousal levels than recorded speech based on the adaptive nature of live speech, we could only confirm this for joyful but not for aggressive voices. A possible explanation might be that aggressive voices are naturally of high arousal levels limiting additional modulation ranges in live speech, while joyful voices might involve more subtle arousal dynamics. Furthermore, the analysis of the dynamic ratings averaged over time reduces information content. Thus, a more detailed exploration of the ratings was incorporated in the correlation analysis (Fig. 5C, and discussed below).

3.3. Acoustic nature of real-time and adaptive affective speech

Several acoustic features related to vocal timbre and pitch, spectral voice variations, and loudness parameters were significantly different in live compared to recorded speech. This seems evolutionarily plausible since affective speech might have evolved to maximize communicative and intended behavioral effects in listeners (Ehret, 2006), and thus its evolutionary benefit could have only been experienced and was only relevant in interactive social settings. Relatedly, neural systems and mechanisms for affect decoding in listeners have also been evolutionarily shaped in interactive and natural settings (Janak and Tye, 2015; LeDoux, 2012; Sokolowski and Corbin, 2012), and, as shown here, speakers could more easily maximize listeners' neural response patterns in interactive experimental settings. We must note that we did not find strong links between speakers' acoustic parameters and listeners' neural dynamics in terms of instantaneous acoustic-to-brain correlations (Fig. 5B). This suggests that natural and adaptive communicative settings might shape neural responses more in terms of long-term adaptation effects based on the general sensitivity and plasticity induced by such live and dynamic communication.

Besides overall acoustic differences, each emotion also revealed specific effects during the live speech condition. Live aggression had a higher vocal pitch range and pitch rise and more vocal power in high-frequency bands. The latter features are very specific for aggressive compared with other affective speech intonations (Patel et al., 2011), and these features seemed even more pronounced in a live and adaptive communication context. Contrarily, live joyful speech specifically had increased bandwidth of the voice formant F3. The voice formant F3 is a specific feature indicating joy compared with other affective speech intonations (Patel et al., 2011; Waaramaa et al., 2006). Additionally, during live affective speech, speakers not only increased certain vocal features but also seemed to suppress or lower other features, such as mean loudness parameters, higher-order MFCCs, spectral flux variability, and lower formant-related parameters. Lowering these features might help to sharpen a distinctive vocal profile that more effectively influences listeners' neural and perceptual responses.

Besides these general acoustic profiles of live aggressive and joyful speech, listeners might differ in their sensitivity to certain vocal feature patterns to accurately decode emotions (Frühholz et al., 2017). In line with this notion, we could establish four principal vocal profiles for both emotions as distinctive prototypes of how speakers can modulate their speech to individually maximize neural and perceptual effects in listeners. We must note that these profiles were not reflecting individual speakers or speakers' gender but were instead found across speakers. The four principal vocal profiles might reflect the different affective subtypes for each of the two emotions expressed by speakers (e.g., hot and cold anger (Bänziger and Scherer, 2005), or amusement and elation (Kamiloğlu et al., 2020)), but it might also reflect the responsiveness of listeners to certain acoustic patterns that individually elicit certain emotions (Banse and Scherer, 1996; Liu et al., 2015). Altogether, the general acoustic profiles of speakers' live affective speech intonations as discussed above and the differential acoustic profiles discussed here confirm that affective speech adapted to listeners' feedback is more distinct and acoustically pronounced than in non-adaptive and recorded speech contexts. This indicates that previous research potentially underestimated the dynamic nature of emotional communication.

3.4. Communicative speaker-listener coupling

Natural communication also involves a co-dependent speaker-listener coupling, and we demonstrate this by correlation analyses between acoustic speaker and neurocognitive listener parameters. In terms of a speech-to-brain coupling, we found couplings between some acoustic speech features and brain activity in specific regions of the higher-order AC. Especially left posterior higher-order AC (i.e., pSTC) activity in listeners correlated with acoustic features of the speakers' affective intonations, particularly with features related to unvoiced speech segments. Unvoiced segments are vocalizations without vibration of the vocal folds and are thus more fricative, hissing, and breathier in acoustic nature. We found that less durational variability of live unvoiced segments correlated with left pSTC activity, such that having a more consistent duration of the UV segments might render live affective speech more authentic (Jürgens et al., 2011). For live aggressive speech, the left pSTC activity was also associated with the F2/F3 formant amplitude and vocal loudness. The pSTC is strongly involved in decoding spectral and intensity features to discriminate affective speech (Frühholz et al., 2016c, 2012), and it probably integrates these features over time (Frühholz et al., 2016b). Regarding the amygdala activity, we did not find correlations with any acoustic features. However, while the amygdala sometimes seems sensitive to discriminative vocal features (Pannese et al., 2016), it seems more susceptible to short acoustic events, but not in tracking acoustic features over time, as in our study.

The latter observation of no amygdala coupling to specific acoustic voice features seems important because the neural patterns we found for live affective speech might have been driven solely by acoustic differences between live and recorded affective speech. Given that we only found very few significant correlations between acoustic speech features and brain activity and given that the left amygdala as the source and target for the real-time brain signal was not directly affected by any acoustic speech features, the neural pattern differences between live and recorded affective speech could have been driven by perceptual differences, indicating that the listeners' subjective experience of live affective speech is different compared to recorded speech.

Besides the speech-to-brain coupling analysis, we therefore also performed speech-to-perception coupling and found extensive relationships between the speakers' affective speech features and the listeners' perceptual ratings. Especially during the live speech condition, we found that listeners' perceptual arousal and valence ratings were negatively associated with voice timbre features (i.e., MFCC features), the strength and variability of UV features, and the variability of formants F1–F3. Contrarily, we found a strong positive association of arousal and valence ratings with the overall strength of voice formants F1–F3 for live speech. These formants usually contribute to a distinctive live vocal tonality and articulation, and a consistent vocal expression along these formants seemed to drive listeners' ratings more than high variability did in these formants and the presence of unvoiced intonations. Concerning each emotion separately, a higher degree of loudness variations influenced higher arousal ratings of live joy. Contrarily, a higher degree of spectral variations influenced the valence ratings and especially the arousal ratings for live aggressive speech. Spectral variations are an important cue to vocal information (Staib and Frühholz, 2020) and more often appear in natural and adaptive vocalizations (Anikin and Lima, 2018). Valence ratings of live aggressive speech were also associated with changes in the alpha ratio, which reflects the relative power of higher-to-lower voice frequency bands and is an important cue to impactful aggressive tones in speech (Patel et al., 2011). Altogether, these speaker-listener parameter associations show that, almost exclusively during live speech, there is a strong relationship between certain features of the speakers' affective intonations and listeners' emotional ratings.

3.5. Methodological considerations

Before summarizing our data, we wanted to discuss a few methodological considerations in our study. First, we chose a visual bar as the feedback signal instead of, for example, dynamic smileys or faces, which are often used in neurofeedback studies (Mathiak et al., 2015; Sitaram et al., 2017). Using social stimuli in neurofeedback studies (e.g., faces with different levels of emotional expressions) introduces a secondary social dimension. Since our study used emotions as the primary social dimension, we aimed to avoid an additional but rather uncontrolled social dimension in our experiment. For our study, a visual bar more directly transmitted an unambiguous and explicit quantification of the listeners' brain signals, which was the central target signal here, thereby simplifying the speakers' task of maximizing the feedback signal. In addition, it ensured that the speakers could fully concentrate on their task instead of focusing on the appearance of the feedback and enabled the most direct and most immediate reaction of the speakers. Second, neutral speech intonations were only included during the perceptual rating of the recorded condition, but not during the real-time neuroimaging and especially not during the live condition. The study's primary focus was on comparing emotionally negative and positive live vocalizations with their recorded counterpart. Unlike affective speech that speakers can modulate to maximize brain effects in listeners, for neutral speech such a task (i.e., maximize "neutrality" in speech intonations) is rather difficult to accomplish. Additionally, the added time participants would have spent in the scanner was not reasonable or feasible. However, the neutral pre-recordings complemented the affective speech in the perceptual rating assessment to provide a reference to what would be considered non-arousing or of no affective quality.

4. Conclusion

Altogether, using a closed-loop neurodynamic setup during affective communication, we here show how speakers adapt their affective speech performance to listeners' dynamic neural activity. Live and adaptive affective speech intonations had a different and distinct acoustic profile compared with pre-recorded speech, as used in many previous studies to define the neural network for vocal affect recognition. Furthermore, live affective speech elicited neural network dynamics in listeners that indicate neural optimization for decoding relevant affective information. Thus, neural affect decoding mechanisms seem more responsive to interactive and individually adaptive communicative contexts and should also be examined in such bi-directionally reactive settings to better capture the full neural processing capacities involved in real-life affective communication.

5. Materials and methods

5.1. Participants ("listeners")

Twenty-four healthy volunteers (14 females, 10 males; mean age 25.42, SD = 3.66) took part in the fMRI experiment as "listeners". The participants listened to affective speech of "speakers" who produced these speech intonations either directly to the participants inside the MR machine ("live") or in a previous speech recording session ("recorded" see below). The listening participants had normal hearing abilities and normal or corrected-to-normal vision. No participant presented a neurological or psychiatric history. All participants gave informed and written consent for their participation in accordance with the ethical and data security guidelines of the University of Zurich. The experiments were approved by the Swiss governmental ethics committee (BASEC Nr. 2017-01086). The participants were monetarily compensated.

5.2. Vocal speakers

We invited eight advanced acting students from the Zurich University of the Arts (4 females, 4 males; mean age 23.63, SD = 1.85) as speakers for the experiment. Through their acting studies, these actors were experienced and trained in expressing and portraying affective states, and they had advanced vocal training. Thus, they represented the ideal speakers for our experimental setup as it was necessary that the speakers were in full control of all their vocal range. Additionally, they were adept at learning text by heart, which was important to cite the text fluently and fully pay attention only to the listeners' amygdala signal.

5.3. Experimental stimuli

As experimental stimuli for the affective speech, we used two so-called pseudo-sentences (e.g., "*Nikalibam sud molen kudsemina lod belam.*", see *SI Appendix*, Table S4 for the full sentences) of 30 s duration that were similar to previously used pseudo-sentences (Frühholz et al., 2015a; Grandjean et al., 2005; Klaas et al., 2015). These pseudo-sentences resemble normal speech but carry no semantic meaning. The sentences missed punctuation to prevent speaking pauses and to allow flexibility in the modulating performance of speakers. This resulted in slightly differing segments of the sentences in each 30-s vocalization period. However, as we purposely chose meaningless words, the focus of the experiment was entirely on the acoustic properties of the speech instead of the content. Listeners were also told that the speech content had now meaning, and that they should carefully listen to the affective intonations. Speakers were asked to speak these pseudo-sentences in a neutral and two affective vocal tones (i.e., aggressive and joyful). These pseudo-sentences were spoken in two different experimental conditions:

(a) Prior to the main fMRI experiment, the speakers were invited to a speech recording session in an anechoic recording chamber to speak these sentences in each vocal tone (referred to as the "recorded" condition).
(b) During the main fMRI experiment, the same speakers were invited to speak the same pseudo-sentences in both affective tones into a microphone that was directly linked to the headphone of listeners inside the fMRI machine (referred to as the "live" condition); the microphone was positioned in an anechoic chamber located near the MR scanner and connected to the experimental computer via cable connections.

The speech recordings were 16-bit recordings with a sampling rate of 44.1 kHz. All pre-recorded affective speech intonations were cut to an exact 30-s duration and were mean RMS normalized to an average sound intensity of approximately 70 dB SPL. For each speaker, we had 18 pre-recorded affective speech periods, resulting from three different utterances for each of the two pseudo-sentences and for each of the three affective intonations. All speech recordings were qualitatively checked for inclusions of a continuous expression of the intended affective voice quality without extended periods of silence or of an unidentifiable affective tone.

5.4. Experimental setup and procedures

The experimental setup included several major parts: (a) The imaging session started with the recording of an anatomical and a few functional images of the listening participant's brain. (b) We then performed a functional voice localizer sequence with the listeners (see below). (c) The anatomical image was used to manually segment the left amygdala of each listening participant with BrainVoyager (v2.8; RRID: SCR_013057; brainvoyager.com/products/brainvoyagerqx.html) by a trained segmenter according to anatomical landmarks, as in several of our previous studies (Frühholz et al., 2015a; Frühholz and Grandjean, 2013). We chose the left amygdala because in previous studies, it showed more reliable and consistent activation in response to affective speech than did the right amygdala (Frühholz et al., 2015a, 2012). (d) The segmented amygdala masks were intra-individually co-registered with the few previously acquired functional brain images. (e) The main experiment comprised eight experimental blocks, during which we continuously sampled brain activity from the whole brain and the left amygdala mask by using real-time fMRI. In addition, the mean left amygdala activity from each scan acquisition was fed back to the speakers in real time via a visual scale. The speakers were located in an anechoic booth near the scanner room (see Fig. 1). (f) Within 2–3 days after the fMRI experiment, we asked the listening participants to perform a post-experimental rating of the speech intonations they had heard during the fMRI experiment (see below).

In the main experimental part (e), each of the eight blocks included an alternating sequence of seven resting periods with no auditory stimulation (each 30 s) and six speech periods (each 30 s). The order of the eight blocks was random and counterbalanced across participants. In four of the eight blocks, participants listened to the pre-recordings ("recorded" condition) and in the other four blocks, to the dynamic live speech of the actors ("live" condition). Both speech conditions for one listening participant were produced by two actors (one male, one female) selected from the total eight speakers. For each participant, speakers for the recorded and the live conditions were the same. While ensuring an equal distribution across all participants, we randomly assigned the actors to the participants.

Thus, the experiment had a factorial design, including the factors *speaking mode* (two levels: real-time "live" condition, pre-recorded "recorded" condition) and *emotional tone* (two levels: aggressive, joyful). We introduced the factor *speaking mode* because we are interested in the influence of dynamic and individually adapted affective communication (which is an ecologically more valid design feature in relation to many real-life and "live" settings) in comparison to affective speech that is not adapted to a listener's reaction, as it was done in many previous studies in the research field (Frühholz et al., 2016b; Frühholz and Schweinberger, 2021). The latter was frequently used in previous studies by using recorded affective speech during the experiment (Frühholz et al., 2016b).

In general, in real-time fMRI neurofeedback studies where brain activity is regulated by the participants inside the scanner themselves, an appropriate baseline condition needs to be chosen as a comparison to the blocks where the brain signal is actively modulated. A rarely used baseline condition is using an experimentally unrelated control region for quantifying a random neurofeedback signal. While we assessed this possibility in a piloting experiment, the randomness of the neurofeedback signal resulted in confusion for the speakers regarding the task to modulate their voice in real time. During voice modulations adapted to the neurofeedback of the target region (i.e., the left amygdala), speakers learned appropriate strategies to maximize the neural signal in each individual listener (see section "Task of the actors"). However, those strategies were not applicable for the control regions and the random feedback thus led to confusions and the unlearning of the acquired strategies. More commonly, studies compare regulation against non-regulation blocks. Therefore, we chose two conditions that differed in the possibility for the speakers to dynamically modulate their speech intonations and to regulate the listening participant's amygdala activity. The "recorded" condition thus served as the baseline condition where no regulation of and adaption to the listener's brain signal was possible. Apart from the non-adaptive nature, however, the recorded condition was similar to the live condition because it included the same speaker, the same expressed emotions, and modulation of vocal features; the latter, however, was introduced in response to an imagined rather than a real social interactor.

5.5. Task of the participants ("listeners") during the fMRI experiment

We told listeners that they will hear emotional voices, and that they should attentively listen to the emotional quality of the voice. The listening participants were unaware of the live and recorded conditions, and they were also unaware that their left amygdala activity was being modulated during the live conditions by modulations of the speaker's speech. To assess whether participants attentively listened to each speech period, we asked questions after each trial ("Which emotion did you hear?", "Was it a male or female voice?") that ensured that they were focused on listening to the speech intonations.

5.6. Task of the actors ("speakers") before and during the fMRI experiment

Actors were asked to upregulate the activity of the left amygdala of listeners by dynamically modulating their speech intonations according to several different features, such that they introduced vocal variations that potentially drive a signal increase in listeners' amygdalae. The amygdala activity was represented by visual analog feedback in the form of a bar graph, and the task was to increase the bar in the positive direction as much as possible.

The actors were extensively trained before the experiment on emotional intonations and how specific emotions prototypically sound like. The actors also learned the sentences before the experiment to enable free and fluent speech performance in the live condition. With the training we ensured that actors would not simply introduce significant differences in speech rate and voice intensity (i.e., loudness), but rather to use their full vocal potential and introduce vocal modulations along several dimensions of the voice acoustic space (e.g., timbre, pitch, harmonics-to-noise ratio as well as vocal variations on these features). The actors were also asked to imagine specific situations for each emotional tone (e.g., joyful: reminisce about a happy memory, proclaim the adoration for something, etc.; aggressive: scold someone, be irritated about an annoying person) during vocalizations. Using this repertoire of vocal and mental strategies they were asked to elicit an increase in the amygdala signal along the emotional tone expressed. Since each listeners' amygdala reacted slightly different, actors were asked to learn the best strategy for each listener within the constraints of the target emotions and vocal strategies. The ability of actors to modulate their voice along the target emotions was quantified by the post-experimental ratings and the acoustic voice analyses.

For the speech pre-recordings, actors were instructed to speak to an imaginary other person and to modulate their speech to imagined responses of the person. Although these speech intonations would be acted and not deeply "felt" by these actors, we have recently shown that these "acted" vocal emotions can be classified as being like the target emotion and include bodily responses in speakers that resemble real emotions (Frühholz et al., 2015b; Klaas et al., 2015).

5.7. Functional voice localizer scan

To localize voice-sensitive cortical regions, we used a standard functional voice localizer scan procedure (Pernet et al., 2015). The stimulus material consisted of forty 8-s sound blocks, including 20 human speech and non-speech vocalizations (vocal sounds) and 20 non-human vocalizations and sounds (non- vocal sounds: animal vocalizations, artificial sounds, natural sounds). Sound blocks were presented at approximately 70 dB SPL and in random order separated by 4-s silent breaks.

5.8. Brain image acquisition

We recorded functional imaging data on a 3 T Philips Achieva Scanner equipped with a 32-channel receiver head-coil array and using a T2*-weighted gradient echo-planar imaging (EPI) sequence including parallel imaging (SENSE-factor 2) with the following parameters: 30 sequential axial slices, whole brain, TR/TE: 2 s/30 ms, flip angle = 82°, slice thickness: 3.0 mm, gap: 1.1 mm, field of view (FOV): 240×240mm, acquisition matrix 80 × 80 voxel, resulting voxel size 3 mm^3, axial orientation. Structural images were acquired using a high-resolution magnetization prepared rapid acquisition gradient echo (MPRAGE) T1-weighted sequence and had 1-mm isotropic resolution (TR/TE: 6.73/3.1 ms, voxel size 1 mm^3, 145 slices, axial orientation). Acoustic stimuli were presented via MR-compatible headphones (MR Confon®, Magdeburg, Germany) with a noise-reduction system at an approximate 70 dB SPL both for the recorded and the live condition. Acoustic loudness for the live condition was adjusted for each participant and speaker prior to each experimental acquisition and monitored during the acquisition to keep the volume at an approximately constant level.

5.9. Online functional brain data analysis for real-time fMRI neurofeedback

During the live condition blocks, the left amygdala signal of the listeners was continuously quantified and fed back to the speakers. Real-time fMRI data analysis was performed with Turbo-BrainVoyager (v3.2; RRID:SCR_014175; brainvoyager.com/products/turbobrainvoyager.html).

5.9.1. Statistics

Real-time functional brain data analysis included 3D head motion correction (all volumes are aligned to the first recorded volume), and temporal filtering (drift removal, linear trend confound predictor added to design matrix). For the online functional brain data analysis, we extracted left amygdala activity at one acquisition time-point as the mean signal of all voxels within the manually segmented amygdala mask at this time-point. The feedback signal from the left amygdala was then calculated as a moving average over the current and the two immediately preceding functional images (i.e., the average signal over the last 6 s, i.e., three TRs) and then normalized to a percentage signal change (PSC) by using a reference signal. The reference signal was calculated separately for each 30-s speech period by taking the mean amygdala activity of the last six scans in the immediately preceding resting period prior to each speech period. The reference signal was displayed as the zero point in the middle of the visual scale at the start of each speech period. The visual feedback display was represented by a bar that either increased or decreased from the zero point according to a positive or negative PSC relative to the reference period. The range of the feedback bar was set to +/− 1 PSC, and PSC was calculated online as the percentage of signal increase/decrease in the moving average interval relative to the reference signal.

5.10. Offline functional brain data analysis

Pre-processing and statistical analyses of functional images were performed with the Statistical Parametric Mapping software (SPM12; RRID:SCR_007037; fil.ion.ucl.ac.uk/spm/) implemented in Matlab (2018b; RRID:SCR_001622; ch.mathworks.com/de/products/matlab). Functional data were first manually realigned to the anterior commissure-posterior commissure (AC-PC) axis, with the origin set to the AC. Then we used the spatial-adaptive Non-Local Means (SANLM) denoising filter (Manjón et al., 2010) implemented in the CAT12 toolbox (12.5; RRID:SCR_019184; neuro.uni-jena.de/cat/) to achieve a more robust tissue segmentation result in the structural images, followed by motion correction of the functional images. Each participant's structural image was co-registered to the mean functional image and then segmented with CAT12 to allow estimation of normalization parameters. Using the resulting parameters, we spatially normalized the anatomical and functional images to the MNI stereotactic space. The functional images for the main experiment were resampled into 2 mm^3 voxels. All functional images were spatially smoothed with an 8-mm full-width at half-maximum isotropic Gaussian kernel.

5.10.1. Statistics

For first-level data analysis from the functional voice localizer scan, we used a general linear model (GLM), and vocal and non-vocal trials were separately modeled with a boxcar function aligned to the onset of each stimulus and a duration of 8 s, which was then convolved with a standard hemodynamic response function (HRF). We accounted for serial correlations in the fMRI time series by an autocorrelation function of the order one. Motion parameters, as estimated during the realignment pre-processing step, were entered into the design as regressors of no interest. Contrast images for vocal and non-vocal trials were then taken to separate random-effects group-level flexible factorial analyses (including the factors *subject* and *condition*) in order to determine voice-sensitive regions in both hemispheres of the AC by comparing vocal against non-vocal trials.

The first-level and second-level analyses for the main experiment were identical to those of the functional voice localizer scan, except that we modeled the brain data in the first-level analysis by using the finite impulse response (FIR) model. Using this FIR model, each of the 30-s periods were modeled with a model order of 15 (corresponding to 15 * TR = 30 s) and a window length of 30 s. This FIR modeling allowed us to better account for signal variation in the BOLD (blood-oxygen-level-dependent) signal during the 30-s speech period, which was highly expected, especially during the live condition. Contrast images were created for each of the four conditions by estimating the signal across all 15 time bins for each condition. The MNI coordinates of all significant peak voxels are listed in *SI Appendix*, Table S1.

All contrast images for the main experiment and the functional voice localizer scan were thresholded at a combined voxel-level threshold of $p < 0.005$ and a cluster extent threshold of $k > 46$; The combined voxel- and cluster-level threshold resulted in a threshold of $p < 0.05$ corrected at the cluster level. This cluster-corrected threshold was determined by using the 3DClustSim algorithm implemented in the AFNI software (AFNI_18.3.01; RRID:SCR_005927; afni.nimh.nih.gov/afni; including the new (spatial) autocorrelation function extension) according to the estimated smoothness of the residual images (estimated using the 3dFWHMx AFNI function).

5.11. Granger causality analysis to estimate directional functional brain connectivity

To estimate the directional functional connectivity of the amygdala and other ROIs that were activated during the experiment, we conducted a Granger causality analysis. This analysis determines how the time course in one brain region A can explain the time course in another brain area B. If this determination is significant, we assume that region A is directionally connected to region B. GCA was performed with the MVGC Multivariate Granger Causality Matlab toolbox (Barnett and Seth, 2014) (mvgc_v1.0; RRID:SCR_015755; users.sussex.ac.uk/~lionelb/MVGC/).

We defined the ROIs based on activations resulting from group-level contrasts between the live and recorded speech conditions across and within the aggressive and joyful speech conditions. We thus included 16 ROIs in total from the AC (bilateral Heschl's gyrus, anterior and posterior STC, and right posterior STS), right fOP, right PMC, right IPS, bilateral dStr and vStr, and bilateral amygdalae. The exact MNI coordinates of the peak voxels of each ROI are marked with an asterisk in *SI Appendix*, Table S1. Cortical ROIs comprised voxels within a 3-mm sphere around the peak location. The ROI definition for the subcortical regions was slightly different to ensure that the masks contain only gray matter. For the ROIs in the striatum, we chose a compromise between the different peak coordinates to ensure that peaks were located in the gray matter. For the amygdalae we used the bilateral masks from the AAL2 atlas (Tzourio-Mazoyer et al., 2002) (RRID:SCR_003550; gin.cnrs.fr/en/tools/aal/) as ROIs to ensure a proper coverage of the amygdala regions.

5.11.1. Statistics

From each ROI, we extracted the time course of activity as the first eigenvariate across voxels within the ROI. For each participant, we epoched the ROI time course data for each trial in a time window from speech onset to 44 s after onset to account for any delays in the BOLD response and its relaxation to baseline in each ROI. We pooled all epochs across all participants instead of estimating the GCA for each participant separately. This is in accordance with previous fMRI studies (Frühholz et al., 2016a; Wang et al., 2013), since the number of trials for each participant is usually too low to provide an accurate estimation of brain connectivity. However, a random effects approach would allow to include interindividual variability which presents a limitation in the fixed effects approach we used. Yet, given the existing data, there was no sufficient number of single trials for each participant to use a random effects approach. To mitigate this limitation, we additionally introduced a very conservative significance threshold to reduce the possibility for false positive results. We used F-tests to check for significance of the connections at $p < 0.01$ (FDR-corrected). For each of the four experimental conditions, we ran a separate GCA. To compare the speaking modes, we subtracted the GCA results of the live from the recorded speech condition for each emotion separately. *SI Appendix*, Table S2 provides a full overview of the connectivity parameters between all ROIs.

5.12. Post-experimental perceptual ratings: listeners' task and analysis

Within 2–3 days after the main fMRI experiment, we asked listening participants to dynamically rate the live and recorded affective speech intonations heard inside the scanner during the fMRI experiment. Participants were still unaware of the different speaking conditions. They also rated neutral intonations that we recorded from each speaker in the speech pre-recording session. Listening participants performed the continuous ratings twice on each of the 30-s speech periods, with one rating along the dimension *arousal* (i.e., low to high arousing; level of emotional engagement and intensity of own emotional reaction to vocalizations) and one along the dimension *valence* (i.e., negative versus positive valence; level of perceived negativity or positivity of the vocalizations), as was previously done in our recent study on music listening (Trost et al., 2015). As in our previous study (Trost et al., 2015), we expected that arousal and valence ratings on affective speech could be rated accurately and independently. The participants listened to the speech recordings through high-quality headphones and rated the speech periods by continuously moving a joystick, that was presented on a computer screen, up and down (arousal, from 0 to 1) or left and right (valence, from −1 to +1). The starting point for each 30-s sample was always set to the midpoint of the scale. The ratings were sampled at a frequency of 16 Hz.

5.12.1. Statistics

To test for differences between the conditions and find out if there was an effect of sex of the speaker, we first averaged the rating values over the 4- to 30-s interval of each speech period. The means of each listener were then subjected to a two-way rmANOVA with the two within-subject factors condition (neutral, joyful recorded, joyful live, aggressive recorded, aggressive live) and sex (male, female). For a measure of effect size, we calculated the eta squared (η^2) which describes the part of variance explained by the respective factor. The two-sided post hoc tests were Bonferroni-corrected for multiple comparisons. P-values < 0.05 were considered statistically significant for both the rmANOVA and the post hoc tests. We applied Mauchly's test to determine if the assumption of compound asymmetry was met. Significance of Mauchly's test indicates that sphericity is not met and the estimate of sphericity ε indicates which adjustment to use ($\varepsilon < 0.75$ Greenhouse-Geisser, $\varepsilon > 0.75$ Huynh-Feldt (Girden, 1992)) to correct for violations of sphericity. For both rating dimensions, Mauchly's test indicated that the assumption of sphericity had been violated (valence: $\chi^2(44) = 144.46, p = 1.3 \times 10^{-12}$; arousal: $\chi^2(44) = 130.84, p = 1.5 \times 10^{-10}$), therefore degrees of freedom were corrected using Greenhouse-Geisser estimates of sphericity (valence: $\varepsilon = 0.352$; arousal: $\varepsilon = 0.309$).

5.13. Acoustic analysis of live and recorded affective speech

We analyzed the live and recorded affective speech intonations from the main experiment and the neutral intonations from the speech pre-recordings by using the openSMILE toolbox (Eyben et al., 2010) (2.3; RRID:SCR_021273; audeering.com/opensmile) which extracted acoustic features that are central to the analysis of affective speech (Eyben et al., 2016). We extracted frequency-related parameters (e.g., pitch, jitter), energy/amplitude-related parameters (e.g., loudness, shimmer), spectral (balance) parameters (e.g., alpha ratio, Hammarberg Index, Mel-frequency cepstral coefficient, spectral flux), and temporal features (e.g., rate of loudness peaks, voiced segments). For all parameters, we extracted the mean and the standard variation, resulting in a total of 88 acoustic voice features. *SI Appendix*, Table S3 provides a full list of all included features. The time courses of features were extracted for the 30-s speech periods in time bins of 2 s in correspondence with the sampling rate of brain signals during the fMRI experiment (TR = 2 s).

5.13.1. Statistics

Across the 88 acoustic features extracted from each speech period, we also calculated difference scores between the live and recorded condition for each participant. These difference scores were subjected to two different analyses. First, we tested for statistical differences across all participants by applying two-sided one-sample t-tests (difference score against 0) for each of the 88 features ($p < 0.0001$, FDR-corrected). Second, we performed a principal component analysis with varimax rotation on the acoustic difference score profile of each participant. This was done with the aim to identify major acoustic vocal profiles of speakers that influence the perceptual and brain data of listeners.

5.14. Correlation analyses between speaker and listener data

The time courses of vocal features were subjected to correlation analyses with the time courses of both perceptual rating dimensions and with the time courses of brain data in different regions (i.e., the same 16 ROIs as defined in the GCA). We downsampled the perceptual ratings to 0.5 Hz so that they would correspond to the sampling rate of the functional brain data (TR) and acoustic features. To evaluate the relationship between vocal feature and brain activity, we shifted the vocal features by 4 s (i.e., 2 *TR) to account for the delay of the BOLD signal (Baumann et al., 2010).

5.14.1. Statistics

We estimated the Pearson correlation coefficient of each acoustic feature to perceptual rating or brain data relationship for each single 30-s speech period of the main experiment for each participant separately. This resulted in correlation coefficient distributions containing 288 coefficients for each acoustic feature within each of the four experimental conditions across all listening subjects (24 (subjects) × 12 (speech periods per condition)). For each of the four experimental conditions, the vocal feature–brain data relationships resulted in 1′408 distributions (88 (vocal features) × 16 (ROIs)) and the vocal feature–rating relationships resulted in 176 distributions (88 (vocal features) × 2 (rating dimensions)).

The correlation coefficients were transformed by using the Fisher-z-transformation procedure, and the distribution of these transformed coefficients were fitted with a normal distribution, resulting in two fit parameters (M, SD). To test whether the distribution of correlation coefficients was significantly different from a normal distribution with M = 0, we performed a permutation test to acquire a null distribution with an estimated mean and standard deviation. For the permutation, we shuffled the acoustic features (n = 88) to speech period (n = 12) relationship 500 times for each condition and each participant, resulting in 101,376 permutations for the entire sample. Each fitted distribution of the empirical data (empirical M and SD) was compared with the normal distribution resulting from the permutation sampling (permutation M and SD) by using two-sided Kolmogorov-Smirnov tests; significance was tested for each of the 88 acoustic features in each condition by using an FDR-corrected level of $p < 10^{-11}$. The same procedure as described above was also applied to the relationships between the time courses of the perceptual ratings and the brain data in all 16 ROIs, resulting in 32 correlation coefficient distributions (2 (rating dimensions) × 16 (ROIs)) (*SI Appendix*, Fig. S4).

Author contributions

F.S. and S.F. contributed to designing the experiments, data acquisition, data analysis, and writing the manuscript; J.D. and P.S. contributed to data acquisition; A.R. contributed to designing the experiment and writing the manuscript; N.F. and E.S. contributed to writing the manuscript.

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Competing interest statement

The authors declare no competing interests.

Data availability

According to the Swiss Human Research Act, we cannot openly share participants' and actors' experimental acoustic and brain data without the explicit agreement of study participants. We only have an agreement by a part of the participants and no agreement by any of the actors and are thus not allowed to publicly share all data supporting our findings. However, data can be shared upon reasonable requests to the corresponding authors in consultation with the local ethics committee of the Cantone Zurich. Regarding data from the group level analysis, the group T-maps of the fMRI data are available on NeuroVault (identifiers.org/neurovault.collection:9810) and the MNI coordinates of all significant peak voxels are listed in Table S1. For the GCA, the resulting group-level connectivity parameters between all ROIs are listed in Table S2.

Code availability

No custom-made code was used for the analyses in this study. We used standard codes implemented in the Statistical Parametric Mapping software (SPM12; RRID:SCR_007037; fil.ion.ucl.ac.uk/spm/) for analysis of the fMRI data. For the Granger causality analysis, we used the MVGC Multivariate Granger Causality Matlab toolbox (mvgc_v1.0; RRID:SCR_015755; users.sussex.ac.uk/~lionelb/MVGC/) with default settings. The perceptual rating, acoustic feature, and correlation analyses were computed with standard Matlab functions (2018b; RRID: SCR_001622; ch.mathworks.com/de/products/matlab) for all statistical steps that are extensively detailed in the Methods.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.pneurobio.2022.102278.

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